

→ Log-Likelihood of Dataset Given HMM

Let T be the total # of timepoints in a sequence and $x_{t:r}$ be the observations from timepoint t to timepoint r . Then the likelihood of the sequence given the model is

$$\mathcal{L} = p(x_{1:T}) = p(x_T, x_{1:T-1}) = p(x_T | x_{1:T-1}) p(x_{1:T-1}) = p(x_T | x_{1:T-1}) p(x_{T-1} | x_{1:T-2}) \dots p(x_1) = \prod_{t=1}^T p(x_t | x_{1:t-1}), \text{ and the log-likelihood is}$$

$$\log \mathcal{L} = \log p(x_{1:T}) = \sum_{t=1}^T \log p(x_t | x_{1:t-1})$$

Let s_t denote the state at time t and $\underline{A}$ be the $k \times k$ state transition matrix learned by the model. Then

$$\begin{aligned} p(x_t | x_{1:t-1}) &= \sum_j p(x_t, s_t = j | x_{1:t-1}) \\ &= \sum_j p(x_t | s_t = j) p(s_t = j | x_{1:t-1}) \\ &= \sum_j p(x_t | s_t = j) \left(\sum_i p(s_t = j | s_{t-1} = i) p(s_{t-1} = i | x_{1:t-1}) \right) \\ &= \sum_j p(x_t | s_t = j) \left(\sum_i \underline{A}_{i,j} p(s_{t-1} = i | x_{1:t-1}) \right) \end{aligned}$$

$$\begin{aligned} p(s_t = j | x_{1:t}) &= \frac{p(s_t = j, x_{1:t})}{p(x_{1:t})} \\ &= \frac{p(x_t, s_t = j, x_{1:t-1})}{p(x_t, x_{1:t-1})} \\ &= \frac{p(x_t | s_t = j, x_{1:t-1}) p(s_t = j, x_{1:t-1})}{p(x_t | x_{1:t-1}) p(s_t = j | x_{1:t-1})} \\ &= \frac{p(x_t | s_t = j, x_{1:t-1}) p(s_t = j | x_{1:t-1})}{p(x_t | x_{1:t-1}) p(s_t = j | x_{1:t-1})} \\ &= \frac{p(x_t | s_t = j, x_{1:t-1}) p(s_t = j | x_{1:t-1})}{p(x_t | x_{1:t-1}) p(s_t = j | x_{1:t-1})} \end{aligned}$$

$$\begin{aligned} \log p(x_t | x_{1:t-1}) &= \log \sum_j p(x_t, s_t = j | x_{1:t-1}) \\ \text{*\log-c} &= \log \sum_j \exp(\log p(x_t, s_t = j | x_{1:t-1})) \\ &\quad \text{*\log-fit (unnormalized)} \end{aligned}$$

$$\begin{aligned} &= \log \sum_j \exp(\log p(x_t | s_t = j) p(s_t = j | x_{1:t-1})) \\ &= \log \sum_j \exp(\log p(x_t | s_t = j) + \log p(s_t = j | x_{1:t-1})) \\ &\quad \text{*\log-B} \quad \text{*\log-pred} \end{aligned}$$

$\log p(x_t | s_t = j)$ is the emission log-likelihood for one timepoint t . `osld` provides a function to compute this

$$\begin{aligned} \log p(s_t = j | x_{1:t-1}) &= \log \sum_i \underline{A}_{i,j} p(s_{t-1} = i | x_{1:t-1}) \\ &= \log \sum_i \exp(\log \underline{A}_{i,j} + \log p(s_{t-1} = i | x_{1:t-1})) \end{aligned}$$

$\underline{A}_{i,j}$ is learned by the model

$$\log p(s_{t-1} = i | x_{1:t-1}) \text{ comes from the previous iteration, so for each time } t, \text{ we must carry forward } \log p(s_t = i | x_{1:t})$$

*\log-fit (normalized)

$$\begin{aligned} \log p(s_t = i | x_{1:t}) &= \log \frac{p(x_t, s_t = i | x_{1:t-1})}{p(x_t | x_{1:t-1})} \text{ by *} \\ &= \log p(x_t, s_t = i | x_{1:t-1}) - \log p(x_t | x_{1:t-1}) \\ &= \log\text{-fit(unnormalized)} - \log\text{-c} \end{aligned}$$

Note: For $t=1$, $\log\text{-c} = \log p(x_1)$ because there's nothing to condition on yet. Similarly, $\log\text{-pred} = \log(s_1 = j) = \pi_j$, the j th element of the initial state distr. $\pi \in \mathbb{R}^k$

Thus, the full log-likelihood of a dataset $\mathcal{D}$ given a model, $\log \mathcal{L}(\mathcal{D})$, can be solved for iteratively. Suppose there are b batches in the dataset, each containing q sequences. Then

$$\mathcal{L}(\mathcal{D}) = \prod_{u=1}^b \prod_{v=1}^q \mathcal{L}(x_{u,v}), \text{ where } x_{u,v} \text{ is the } v^{\text{th}} \text{ sequence of batch } u, 1 \leq u \leq b, 1 \leq v \leq q. \text{ So}$$

$$\log \mathcal{L}(\mathcal{D}) = \log \prod_{u=1}^b \prod_{v=1}^q \mathcal{L}(x_{u,v}) = \sum_{u=1}^b \log \prod_{v=1}^q \mathcal{L}(x_{u,v}) = \sum_{u=1}^b \sum_{v=1}^q \log \mathcal{L}(x_{u,v}), \text{ i.e. the sum of the log-likelihoods of all } b \cdot q \text{ sequences}$$

*\full_loglik